TRACE: Target Reliable Adaptive CLIP Enhancement via LoRA for Source-Free Domain Adaptation

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I. Theoretical Analysis

We provide a theoretical motivation for two key designs of TRACE: cross-model agreement for reliable target supervision and frozen-CLIP anchoring for constrained guidance tuning.

Motivation 1 *Let $x \in \mathcal{D}_t$ be a target sample with ground-truth label $y \in \{1, \ldots, K\}$. Let $\hat{y}^s$ and $\hat{y}^0$ denote the predictions of the source model and frozen CLIP, whose target domain error rates are $\epsilon_s = \Pr(\hat{y}^s \neq y)$ and $\epsilon_0 = \Pr(\hat{y}^0 \neq y)$, respectively. Assume that, conditioned on making an error, each predictor assigns x uniformly to one of the $K-1$ incorrect classes, and their errors are independent. Then, conditioned on the agreement event $\mathcal{A} = \{\hat{y}^s = \hat{y}^0\}$, the error rate of the agreement-based pseudo label is:*

$$\epsilon_{\mathcal{A}} = \Pr(\hat{y}^s \neq y \mid \mathcal{A}) = \frac{\epsilon_s \epsilon_0/(K-1)}{(1-\epsilon_s)(1-\epsilon_0) + \epsilon_s \epsilon_0/(K-1)}. \tag{1}$$

Moreover, when both predictors are better than random guessing and K is moderately large, $\epsilon_{\mathcal{A}}$ is smaller than the error rate of either individual predictor. Thus, source-CLIP agreement provides a lower-noise supervisory signal for target-aware consistency learning.

This work was supported by the National Natural Science Foundation of China under Grant 62676199 and 62176128, the Basic Research Program of Jiangsu under Grant BK20231143, the Open Project of Key Laboratory of Tibetan Information Processing of the Ministry of Education under Grant QHSF-CS-2609, the Fundamental Research Funds for the Central Universities No. NJ2023032, as well as the 333 High-Level Talent Project of Jiangsu Province. (Corresponding author: Qing Tian.)

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Proof 1 *The probability that the two predictors agree on the correct class is:*

$$\Pr(\hat{y}^s = \hat{y}^0 = y) = (1-\epsilon_s)(1-\epsilon_0). \tag{2}$$

The probability that they agree on the same wrong class is:

$$\Pr(\hat{y}^s = \hat{y}^0 \neq y) = \sum_{c \neq y} \Pr(\hat{y}^s = c, \hat{y}^0 = c) = \frac{\epsilon_s \epsilon_0}{K-1}. \tag{3}$$

Therefore, the correctness probability under agreement is:

$$\Pr(\hat{y}^s = y \mid \mathcal{A}) = \frac{(1-\epsilon_s)(1-\epsilon_0)}{(1-\epsilon_s)(1-\epsilon_0) + \frac{\epsilon_s \epsilon_0}{K-1}}. \tag{4}$$

Equivalently, it can be deduced Eq. 1. This expression shows that the probability of consistent wrong prediction is suppressed by the factor $K-1$. Thus, when the source model and frozen CLIP make the same prediction, especially in a multi-class setting, their consensus is substantially less noisy than a single prediction source. In TRACE, we further require both predictors to be confident:

$$\mathcal{A}_\tau = \left\{ \hat{y}^s = \hat{y}^0,\ \max p^s \geq \tau,\ \max p^0 \geq \tau \right\}. \tag{5}$$

Under calibrated confidence, conditioning on $\mathcal{A}_\tau$ further reduces the effective error rates of both predictors. This motivates using high-confidence source-CLIP agreement as positive target guidance.

However, adapting CLIP only with target domain signals may cause semantic drift, since no target labels or source samples are available. TRACE therefore constrains the tuned CLIP branch around the frozen CLIP reference.

Motivation 2 *Let p^0 and p^θ be the prediction distributions of frozen CLIP and the tuned CLIP branch, and let z^0 and z^θ be their normalized image representations. Ideally, guidance learning should solve:*

$$\begin{aligned} \min_\theta \ & \mathcal{L}_{guide}(\theta) \\ & + \lambda_{anc} \mathbb{E}_{x \sim \mathcal{D}_t} \left[\mathrm{KL}(p^0 \| p^\theta) + \eta(1 - \cos(z^0, z^\theta)) \right], \end{aligned} \tag{6}$$

$$\text{s.t.} \quad \mathbb{E}_{x \sim \mathcal{D}_t} \left[\mathrm{KL}(p^0 \| p^\theta) + \eta(1 - \cos(z^0, z^\theta)) \right] \leq \rho,$$

where ρ controls the allowed deviation from frozen CLIP.

Remark 1 *Theorem 2 suggests that the prediction-level KL term bounds the distributional deviation between frozen and tuned CLIP. By Pinsker's inequality:*

$$\|p^\theta - p^0\|_1 \leq \sqrt{2\mathrm{KL}(p^0 \| p^\theta)}, \tag{7}$$

which indicates that minimizing the KL anchor prevents the tuned guidance from moving excessively away from the

Algorithm 1 Training Procedure of TRACE

Require: Source-trained model $h_s = \{f_s, b_s, g_s\}$; unlabeled target data $\mathcal{D}_t$; frozen CLIP model $\mathcal{C} = \{E_v, E_t\}$; class names $\mathcal{Y}$; maximum epochs $E_{\max}$.
Ensure: Adapted target model h_t.

```
// ==== Preparation ====
Initialize target model h_t = {f_t, b_t, g_s} from h_s.
Freeze the classifier g_s.
Construct text prompts and obtain textual prototypes.
Insert LoRA modules into the query and value projections of the last Transformer blocks in E_v.
Freeze CLIP parameters and keep only LoRA trainable.
Initialize the prediction bank S ∈ R^{N_t × K} using h_t on D_t.
// ==== Training Loop ====
for epoch = 1 to E_max do
    for each mini-batch B sampled from D_t do
        // ---- Target-aware CLIP guidance learning ----
        Compute source model prediction p_i^s.
        Disable LoRA and compute frozen-CLIP prediction p_i^0 and feature z_i^0.
        Enable LoRA and compute tuned-CLIP prediction p_i^θ and feature z_i^θ.
        Determine reliable agreement samples by Eq. 10.
        Select complementary classes for N_i.
        Compute target-aware consistency L_tac by Eq. 12.
        Compute frozen-CLIP anchoring by Eq. 13.
        Update LoRA parameters by minimizing Eq. 14.
        // ---- Target model adaptation with TRACE guidance ----
        Recompute current CLIP guidance p_i^θ using the updated LoRA branch.
        Construct mixed pseudo labels ỹ_i using the prediction bank S and p_i^θ.
        Compute adaptation objective L_target by Eq. 19.
        Update f_t and b_t by minimizing Eq. 20.
        Update prediction bank S ← sg[q_i].
    end for
end for
return Adapted target model h_t.
```

frozen CLIP semantic distribution. Meanwhile, the feature-level cosine constraint limits representation drift in the CLIP embedding space.

Together, these two constraints cast CLIP tuning as a trust-region optimization problem: TRACE allows CLIP guidance to absorb reliable target evidence, while preserving the pretrained semantic prior of frozen CLIP.

II. Algorithm

The whole training procedure of TRACE in **Algorithm** 1.